Empowerment Outer-Policy Experiments: Separating Inner Valuation from Outer Control

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Abstract

Empowerment—the channel capacity of an agent's action-to-future-state channel—is a widely studied intrinsic motivation signal. However, prior work often conflates the *empowerment objective* (which states are valued) with the *outer control rule* (how actions are selected given empowerment estimates). This repository provides a controlled comparison: we hold empowerment computation fixed and vary only the outer policy. We compare greedy (argmax), softmax (Boltzmann), and ϵ-greedy outer policies on exact tabular empowerment in a corridor gridworld. Greedy control collapses into stereotyped local loops, while moderate-temperature softmax control escapes local empowerment maxima and reaches higher-empowerment regions at the cost of reduced instantaneous empowerment.

1 Introduction

Empowerment [1] provides an agent-centric, task-independent measure of how much control an agent has over its future. It has been used to explain and generate life-like behaviors such as obstacle avoidance, resource acquisition, and survival [2, 3, 4].

A subtle but important design choice in empowerment-driven agents is the *outer policy*: given empowerment estimates for all reachable states, how should the agent select its next action? The simplest choice is greedy (argmax) over successor-state empowerment. But stochastic alternatives—softmax or ϵ-greedy—can produce qualitatively different behavior even when the underlying empowerment values are identical.

This repository isolates this variable. The central experimental principle is:

Fix empowerment. Vary only the outer policy. Measure behavior broadly.

2 Mathematical Framework

2.1 Empowerment

Consider a discrete environment with state space $\mathcal{S}$ and action space $\mathcal{A}(s)$ available at each state s. The transition dynamics are $P(s' \mid s, a)$.

Definition 1 (n-step empowerment)**.** *For state s and horizon n, the* empowerment *is*

$$\mathcal{E}_n(s) \;=\; \max_{q(a_{0:n-1}|s)} I(A_{0:n-1};\, S_n \mid s) \tag{1}$$

where $I(\cdot;\cdot)$ is mutual information, $A_{0:n-1}$ is a length-n action sequence drawn from the input distribution $q(a_{0:n-1} \mid s)$, and S_n is the resulting final state under the environment dynamics.

Empowerment is the *channel capacity* of the n-step action-to-state channel induced by the dynamics at s. It measures, in nats, the maximum number of distinguishable futures the agent can steer toward.

2.2 The Action-to-State Channel

For a given state s and horizon n, we construct the channel matrix $\mathbf{C} \in \mathbb{R}^{|\mathcal{X}|\times|\mathcal{S}|}$ where $\mathcal{X} = \mathcal{A}(s)^n$ is the set of all length-n action sequences. Each entry is:

$$C_{x,s'} = P(S_n = s' \mid s,\, a_{0:n-1} = x) \tag{2}$$

computed by rolling out the transition probabilities:

$$P(S_n = s' \mid s, x) \;=\; \sum_{s_1,\ldots,s_{n-1}} \prod_{t=0}^{n-1} P(s_{t+1} \mid s_t, a_t) \tag{3}$$

where $s_0 = s$ and $x = (a_0, \ldots, a_{n-1})$. In deterministic environments, each row of $\mathbf{C}$ is a one-hot vector.

2.3 Blahut–Arimoto Algorithm

Channel capacity is computed via the Blahut–Arimoto (BA) algorithm [5]. Starting from a uniform input distribution $q^{(0)}$, BA iterates:

1. Compute the output marginal: $r^{(t)}(s') = \sum_x q^{(t)}(x)\, C_{x,s'}$
2. Compute the per-input KL term: $\phi^{(t)}(x) = \exp\Big(\sum_{s'} C_{x,s'} \ln \frac{C_{x,s'}}{r^{(t)}(s')}\Big)$
3. Update input distribution: $q^{(t+1)}(x) \propto q^{(t)}(x)\, \phi^{(t)}(x)$

Convergence is monitored via the gap between the lower bound $C_L = \ln\big(\sum_x q(x)\phi(x)\big)$ and upper bound $C_U = \ln\big(\max_x \phi(x)\big)$. The algorithm terminates when $C_U - C_L < \epsilon$ (we use $\epsilon = 10^{-8}$).

2.4 Empowerment-Derived Action Scores

Given a precomputed empowerment table $\{\mathcal{E}_n(s)\}_{s\in\mathcal{S}}$, we define the one-step action score:

$$Q_{\text{emp}}(s, a) \;=\; \sum_{s'} P(s' \mid s, a)\, \mathcal{E}_n(s') \tag{4}$$

In deterministic environments this simplifies to $Q_{\text{emp}}(s, a) = \mathcal{E}_n(f(s, a))$ where f is the transition function.

This score is the same for all outer policies—the only thing that differs is how Q_{emp} is converted into action probabilities.

3 Outer Policies

3.1 Greedy (Argmax)

The greedy policy selects uniformly among maximizers:

$$\pi_{\text{greedy}}(a \mid s) \;=\; \frac{\mathbf{1}[a \in \arg\max_b Q_{\text{emp}}(s, b)]}{|\arg\max_b Q_{\text{emp}}(s, b)|} \tag{5}$$

3.2 Softmax (Boltzmann)

The softmax policy assigns probability proportional to exponentiated scores:

$$\pi_\tau(a \mid s) \;=\; \frac{\exp\bigl(Q_{\text{emp}}(s,a)/\tau\bigr)}{\sum_b \exp\bigl(Q_{\text{emp}}(s,b)/\tau\bigr)} \tag{6}$$

where $\tau > 0$ is the temperature parameter. Key limits:

- $\tau \to 0$: recovers the greedy policy
- $\tau \to \infty$: approaches the uniform distribution over actions

For numerical stability, we subtract $\max_b Q_{\text{emp}}(s,b)$ from all scores before exponentiation.

3.3 Epsilon-Greedy

The ϵ-greedy policy serves as a control to distinguish "stochasticity helps" from "Boltzmann-shaped stochasticity helps":

$$\pi_\epsilon(a \mid s) \;=\; \begin{cases} \dfrac{1-\epsilon}{|\arg\max_b Q_{\text{emp}}(s,b)|} + \dfrac{\epsilon}{|\mathcal{A}(s)|} & a \in \arg\max_b Q_{\text{emp}}(s,b) \\[2ex] \dfrac{\epsilon}{|\mathcal{A}(s)|} & \text{otherwise} \end{cases} \tag{7}$$

4 Corridor Environment

The corridor environment is a deterministic gridworld designed to create the simplest possible test of local-maximum escape. It consists of:

- A **local basin**: a 7×4 open room (28 states) with moderate empowerment, containing the start state.
- A **narrow corridor**: 3 cells wide at a single row, connecting the two basins. These states have reduced empowerment because fewer future states are reachable within the horizon.
- A **far basin**: a 7×7 open room (49 states) with higher empowerment due to its larger spatial extent.

The agent has four actions (up, down, left, right) at every state. Moving into a wall results in staying in place. There are no absorbing states.

At horizon $n = 3$, the mean empowerment values are:

- Local basin: 2.631 nats
- Far basin: 2.693 nats

The far basin has higher empowerment because its larger area provides more reachable states within any given horizon. However, to reach it, the agent must traverse corridor states with *lower* empowerment than the local basin—a greedy agent will not voluntarily enter them.

5 Experimental Setup

We precompute the exact n-step empowerment table for all 80 states using BA with horizon $n = 3$. We then run rollouts of length $T = 200$ steps for each outer policy across 100 random seeds.

Independent variables.

- Outer policy: greedy, softmax ($\tau \in \{0.01, 0.03, 0.1, 0.3, 1.0, 3.0\}$), ϵ-greedy ($\epsilon \in \{0.1, 0.3\}$)

Dependent variables.

- Fraction of episodes reaching the far basin
- Mean state visitation entropy (Shannon, in nats)
- Mean number of distinct states visited
- Mean empowerment of visited states

6 Results

Table 1: Corridor experiment results ($n = 3$, $T = 200$, 100 seeds).

Policy	Far Basin %	Entropy	Distinct States	Mean Emp.
Greedy	0.00	1.301	6.0	2.861
Softmax $\tau = 0.01$	0.00	1.311	6.3	2.860
Softmax $\tau = 0.03$	0.00	1.646	9.1	2.853
Softmax $\tau = 0.1$	0.00	2.357	16.7	2.820
Softmax $\tau = 0.3$	0.10	2.961	26.0	2.751
Softmax $\tau = 1.0$	0.29	3.269	33.7	2.682
Softmax $\tau = 3.0$	0.33	3.255	33.6	2.652
ϵ-greedy $\epsilon = 0.1$	0.02	1.516	10.3	2.852
ϵ-greedy $\epsilon = 0.3$	0.25	2.027	16.9	2.831

The results in Table 1 support the core hypotheses:

H1: Greedy empowerment is behaviorally narrow. The greedy agent visits only 6 distinct states out of 80 and never reaches the far basin. It becomes trapped in a small loop near the highest-empowerment states of the local basin.

H2: Softmax improves basin escape. At $\tau = 0.3$, 10% of episodes reach the far basin; at $\tau = 1.0$, this rises to 29%. The softmax policy's willingness to occasionally accept lower-empowerment actions allows it to traverse the corridor.

H3: Diversity–empowerment trade-off. Visitation entropy increases monotonically from 1.30 nats (greedy) to 3.27 nats ($\tau = 1.0$), while mean empowerment of visited states decreases from 2.86 nats to 2.68 nats. Softmax explores more broadly but spends less time at peak-empowerment states.

H4: Diminishing returns at high temperature. Between $\tau = 1.0$ and $\tau = 3.0$, the far-basin reach rate plateaus (29% $\rightarrow$ 33%) while entropy saturates and mean empowerment continues to decline. Very high temperatures approach uniform random behavior.

Boltzmann vs. uniform stochasticity. At comparable diversity levels, ϵ-greedy with $\epsilon = 0.3$ reaches the far basin 25% of the time but visits only 16.9 distinct states, whereas softmax at $\tau = 1.0$ reaches 29% and visits 33.7 distinct states. The Boltzmann weighting distributes exploration probability more effectively than uniform noise.

7 Repository Structure

The implementation follows a strict separation between inner empowerment computation and outer policy selection.

`src/envs/` Environment interface (`DiscreteEnv` base class) and the corridor gridworld.

`src/empowerment/` Exact empowerment computation: `nstep_model.py` builds the n-step channel matrix, `blahut_arimoto.py` computes channel capacity, and `exact.py` ties them together. `score.py` computes $Q_{\text{emp}}(s, a)$.

`src/policies/` Outer policies (greedy, softmax, ϵ-greedy), all consuming the same Q_{emp} scores.

`src/rollout/` Seeded simulation loop.

`src/metrics/` Visitation entropy, distinct states, basin-reaching fraction.

`scripts/` Experiment runner with temperature sweep and summary table output.

`tests/` Unit tests for BA convergence, environment dynamics, and policy properties (17 tests).

8 Discussion

These results demonstrate that behavioral conclusions about empowerment-driven agents can depend substantially on the outer control rule. The greedy agent's stereotyped looping is not a property of empowerment *per se*—it is a property of greedy action selection applied to empowerment scores. A softmax agent using the *identical* empowerment values exhibits qualitatively different behavior: broader exploration, basin escape, and higher trajectory diversity.

This distinction matters for interpreting prior work. When a paper reports that "empowerment-driven agents exhibit behavior X," the outer policy is a confound. Our framework makes it possible to ask: is X caused by what empowerment values, or by how the agent acts on those values?

Limitations. The current results use a single environment with a small empowerment gap between basins ($\Delta\mathcal{E} \approx 0.06$ nats). Larger environments, longer horizons, and environments with survival pressure (e.g., four-rooms with energy) would provide stronger tests. The softmax temperature is also environment-specific; score normalization (z-score mode) would improve cross-environment comparability.

Future work. Planned extensions include four-rooms with energy constraints, a noisy room variant, discretized cartpole, and horizon sweep ablations.

References

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